

DiscountCalculator

1. Input Space Partitioning (ISP)

In this project, the Input Space Partitioning (ISP) technique was used to organize the different types of inputs for the system. The system behavior depends on three main characteristics, which were divided into separate blocks.

Customer Type

This characteristic describes the type of customer using the system.

- NEW
- REGULAR
- PREMIUM

Total Orders in Last Year

This characteristic represents the number of orders made by the customer during the previous year.

- LOW (<10 orders)
- HIGH (10 or more orders)

Newsletter Subscription

This characteristic shows whether the customer is subscribed to the newsletter or not.

- TRUE
- FALSE

These blocks are considered complete because they cover all possible cases, and they are disjoint because each input belongs to only one block.

Test Case Customer Type Orders Newsletter Subscription

T1	NEW	LOW	TRUE
T2	NEW	LOW	FALSE
T3	REGULAR	LOW	TRUE
T4	REGULAR	HIGH	FALSE
T5	PREMIUM	LOW	FALSE
T6	PREMIUM	HIGH	TRUE

3. Infeasible Combination

One invalid combination was identified during testing.

A customer with the type NEW cannot logically have 10 or more previous orders. Because of this, the combination between NEW customers and HIGH orders is considered infeasible. In the implementation, the program handles this case by throwing an `IllegalArgumentException`.

4. Limitations of Pairwise Testing

Pairwise testing is useful because it helps reduce the total number of test cases without needing to test every possible combination. It focuses on covering all possible pairs of inputs, which can detect many common defects.

However, pairwise testing still has limitations. Some software defects only appear when three or more variables interact together. Since pairwise testing

only guarantees coverage between pairs of inputs, those complex interaction bugs may not be detected. Because of this, pairwise testing improves efficiency, but it does not guarantee a completely bug-free system.

5. Team Contribution Matrix

1-mohammad alkairi : ISP design, pairwise analysis, and implementation

2-haitham alomar : JUnit testing and debugging

Tareq jarrar: Report formatting and documentation

Mohammad alkhairi /2231384

Haitham alomar /2237366

Tareq jarrar/2236415